PROPOSED TRIAL · ONE SAMPLE, ONE SESSION

Can a computed rectangle be seen?

We designed a DNA origami rectangle from scratch with an open toolchain, and predicted what it should look like. We would like to find out whether it folds and whether the prediction holds.

The ask, in four numbers

One folded sample, deposited on mica, imaged in air. Everything before the microscope we do ourselves and bring with us.

INSTRUMENT TIME	SAMPLES	PARTICLES WANTED	OBJECT SIZE
~3 h	**1** (+2 spare)	**300**	**106 × 66** nm

What the sample is

A Rothemund-style rectangle on the square lattice: a single M13mp18 scaffold strand folded by 220 synthetic staples into a flat sheet roughly 106 × 66 nm and one helix thick, about 2 nm tall.

It is deliberately the most ordinary object in the field. The flat rectangle is the standard calibration sample for origami AFM, so if it does not appear as expected, the fault is ours and not the instrument's.

One caveat worth stating plainly. This is Rothemund-*style*, not the published 2006 staple set. The scaffold routing and every staple were generated by our own toolchain, which is exactly the thing being tested. It should fold; it has never been folded.

PROPERTY	VALUE	NOTE
Scaffold	M13mp18, 7249 nt	6930 nt routed; 319 nt left as an unpaired loop
Staples	220	30–32 nt, standard desalted, 25 nmol scale
Lattice	square, 22 × 320 bp	twist-corrected: one base deleted every 64
Predicted footprint	106.2 × 66.0 nm	from the design geometry, not from simulation
Expected height	~2 nm	single helix layer, dried on mica

Sample preparation

Steps 1 to 4 happen in our lab and arrive with the sample. Step 5 is the only one that needs your bench, and it is the one we would most like corrected: you know the mica literature far better than we do.

01

day before

Stocks

- Folding buffer: 5 mM Tris, 1 mM EDTA, pH 8.0, made as a 10× stock without magnesium.
- 1 M $MgCl_2$ stock, added per reaction so each condition gets its own concentration.
- Scaffold resuspended and quantified by A_{260}.
- Staples pooled from three 96-well plates into a single equimolar mix, then diluted so the pool can be added in one pipetting step.

02

35 min + setup

Folding, as a magnesium series

Nine 50 µL reactions, identical except for magnesium: 5, 7, 9, 11, 13, 15, 17, 19, 20 mM $MgCl_2$. Scaffold at 10 nM, each staple at 50 nM (5× excess).

Anneal: hold 65 °C for 15 min, then ramp 60 → 40 °C at 1 °C/min. The whole thing is about 35 minutes, because this is a two-dimensional object; a 3D one would need the overnight ramp.

Magnesium is screened rather than chosen because the optimum moves with the design and we have no prior for this one.

03

~1.5 h

Gel check, to pick one condition

2% agarose in 0.5× TBE with 11 mM $MgCl_2$, run cold at low voltage, against the bare scaffold as a reference lane.

We are looking for a sharp band running above the scaffold, no smear behind it, and nothing stuck in the well. The best-looking condition becomes the sample that travels; two neighbours come along as spares.

04

~1 h

Purification

PEG precipitation, resuspended in folding buffer at the magnesium the reaction used. This removes the large excess of free staples, which otherwise cover the mica and make counting unreliable.

Spin-filter purification (100 kDa) is our fallback if PEG proves lossy. If you would rather we skipped purification entirely and brought the crude reaction, say so — it is one fewer step to blame.

05

~10 min per sample

Deposition on mica

This is the step we have written from the literature rather than from experience, and the numbers below are a starting point we expect you to change.

- Cleave the top mica layer with tape immediately before use.
- Dilute the sample to roughly 1–2 nM in buffer containing 10 mM $MgCl_2$.
- Place a 10 µL droplet on the fresh surface and let it adsorb for 1–5 min.
- Rinse with about 200 µL ultrapure water to carry off excess salt.
- Dry in a gentle nitrogen stream for 2–3 min.

Magnesium mediates the adhesion, so its concentration trades coverage against aggregation, and the rinse trades salt crystals against washing the sample off. Both are exactly the kind of thing we would rather you set than us.

> AIR OR LIQUID
>
> We have written this for drying and imaging in air, which is the simpler ask and enough to answer our question. If imaging in liquid is easier on your setup, we would prefer it: drying is a known source of flattening, and a height measured dry is not the height of the object in solution.

What we would like imaged

Two scales, for two different purposes.

PURPOSE	SCAN	WHY THAT SIZE
Counting and classifying	2 × 2 µm	At 512 px this is ~4 nm per pixel: enough to tell a well-formed rectangle from a malformed one or an aggregate, and to get many per frame
Measuring dimensions	500 × 500 nm	~1 nm per pixel, so an edge can be located well inside our 10% agreement target

How many particles, and why that number

We are after a fraction — what proportion of objects are well-formed — and the precision of a fraction depends only on the count. At 100 particles, a yield near 70% carries a 95% interval of about ±9%. At 300, that tightens to roughly ±5%.

So: **300 well-separated particles if they come cheaply, 100 as the floor.** Below 100 the number stops being able to distinguish a good fold from a mediocre one, which is the distinction we came for. Dimensions need far fewer — 50 clean particles give a solid mean.

What gets compared afterwards

Every measurement below has a predicted counterpart already computed, which is the entire point of the exercise. Nothing here needs deciding at the microscope; it is listed so the imaging is framed to support it.

OBSERVABLE	PREDICTED	FROM THE IMAGES
Long edge	106.2 NM	MEAN ± SD
Short edge	66.0 NM	MEAN ± SD
Height	~2 NM	MEAN ± SD
Well-formed fraction	NO PRIOR	COUNT / TOTAL
Global flatness	FLAT	VISIBLE TWIST?

THE ONE WE ARE LEAST SURE OF

The square lattice puts crossovers every 8 bp and calls 32 bp three turns, which implies 10.67 bp per turn against B-DNA's relaxed 10.5. Left uncorrected the sheet acquires a right-handed twist. We correct it by deleting one base every 64, which cancels the mismatch exactly on paper. Whether it cancels in practice is visible in an AFM image and nowhere else — a twisted rectangle does not lie flat, and it shows.

Where we would value being told we are wrong

We have built the software side and read the literature. We have not done this at a bench, and the failure modes we cannot see are the ones that matter.

Is the deposition sensible?

Magnesium concentration, incubation time, and whether to rinse at all. This is the step most likely to be wrong in ways we would misread as a folding problem.

Should we bring it folded, or fold it there?

Transporting a folded sample risks it; folding on site costs a thermocycler slot and 35 minutes. We are happy either way, and would rather be told than guess.

Purified or crude?

Free staples are a known nuisance on mica, but purification is another place to lose material. Your call.

Anything about this that is simply naive?

An honest read is more useful to us than a successful image. If the object is a poor choice, or the question is not answerable this way, that is worth knowing before the session rather than after.

What comes back

The raw instrument files as recorded, in their native format rather than exported images, alongside the acquisition settings. We process them with our own pipeline, and the measurements go into an open record with the conditions attached — which magnesium, dried or in liquid, how many particles counted.

If the prediction and the measurement disagree, that disagreement is the result we came for, not a failure to be tidied away. It is the only thing that tells us the toolchain is wrong where it is wrong.

Prepared with an open design toolchain. Every number above is generated from the design, not typed in: the staple count, the magnesium series, the predicted dimensions and the cost all come out of the same source that produced the sequences.

Happy to send the design file, the full staple list, or the simulation setup on request.